

Abstract No.

Comparative genetic analyses of glioblastoma-stem-cell based brain tumor organoids vs glioblastoma stem like cells and glioblastoma tissue

Vivian-Pascal Brandt^{1,}, Johannes Wach², Erdem Güresir², and Heidrun Holland¹*

¹Saxonian Incubator for Clinical Translation (SIKT), University of Leipzig

²Department of Neurosurgery, University Hospital Leipzig

*Presenting author

Glioblastoma (GB) is defined as high grade glioma with an astrocytic differentiation (WHO grade 4). Possibly, glioblastoma stem like cells (GSC) could be responsible for tumor initiation, maintenance, recurrence and cancer therapy resistance. Genetic comparisons of glioblastoma tissue vs GSCs and GSC-based brain tumor organoids (GSC-BTO) are still limited. The aim of this study was the genetic comparison of tumor tissue, GSC and GSC-BTO to assess the genetic integrity of GSC-BTO. Primary tumor was obtained during neurosurgical resection and GSCs were cultivated under serum-free conditions. GSC-based organoids were generated using StemDiff™ Cerebral Organoid Kit. Immunohistochemical characterization of GSCs (e.g. SOX2, Nestin) and GSC-BTO (e.g. GFAP, β -Tubulin) was performed. Genetic analyses were carried out using SNP array (Affymetrix CytoScanHD Array). In previously published data (PMID: 32634135), we have detected a similar genetic profile of tumor tissue vs GSC with minor differences. Here, we could identified previously detected chromosomal aberrations in tumor tissue and GSC for GSC-BTO too (e.g. gain 3q21.3-q29). Predominantly, we identified a comparable genetic profile for GSC-BTO in comparison with GSC or tumor tissue of the respective patient with few additional aberrations in GSC-BTO. One donor have showed more chromosomal aberrations in GSC-BTO vs GSC and tumor tissue (e.g. loss of 1q21.2-q44 and gain 8q11.1-q24.3 in GSC-BTO). Interestingly, in this donor, the cn-LOH aberration for chromosomal region 9p21.3-p21.1 were detected only in GSC-BTO (tumor: -; GSC: loss) and could be an indication for a tumor repair mechanisms (somatic reduplication). Predominantly, our results highlights the reflection of the respective patient specific genetic profile of GSC-BTO vs tumor tissue and GSC. Nevertheless, further investigations are needed to enlarge the knowledge about the genetic integrity of glioblastoma-based tumor organoids.



Supported by SciSerTec

Web: <http://www.scisertec.de>